



Training Program 1: Advanced NLP & Large Language Models (LLMs)

- **M01: Text Preprocessing**
- **M02: Word and Sentence Embeddings**
- **M03: Transformer Architectures**
- **M04: Entity-Centric NLP**
- **M05: Applied NLP Tasks**
- **M06: Multimodal NLP**
- **M07: LLM Fundamentals**
- **M08: Prompt Engineering**
- **M09: Fine-Tuning & PEFT**
- **M10: RAG & Real-World Projects**



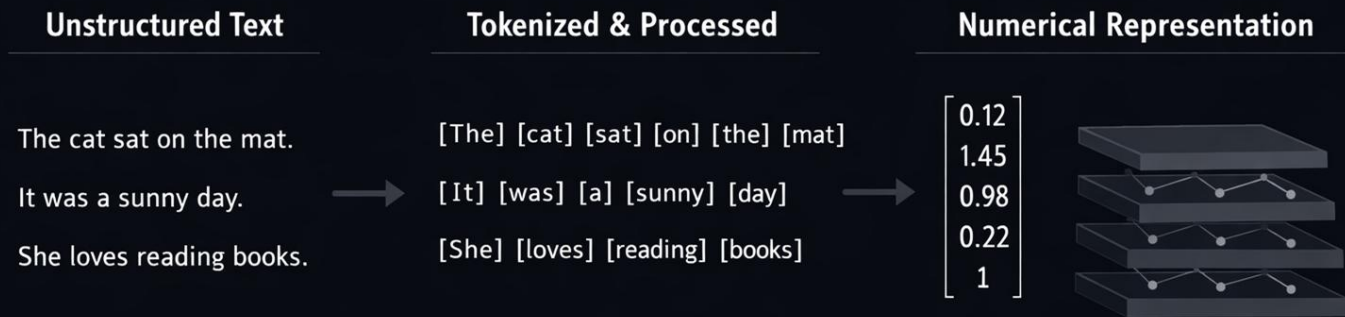
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- **1. Getting Started with Text Processing**
- **2. Fundamentals of Tokenization**
- **3. Byte-Pair Encoding (BPE) and Subword Tokenization**
- **4. Advanced Subword Tokenization with SentencePiece**
- **5. Stopwords and Vocabulary Management**
- **6. Processing Multilingual and Cross-Lingual Text**
- **7. Summary and Next Steps**

M01.1 BPE

Chapter 1.1 – Getting Started with Text Processing

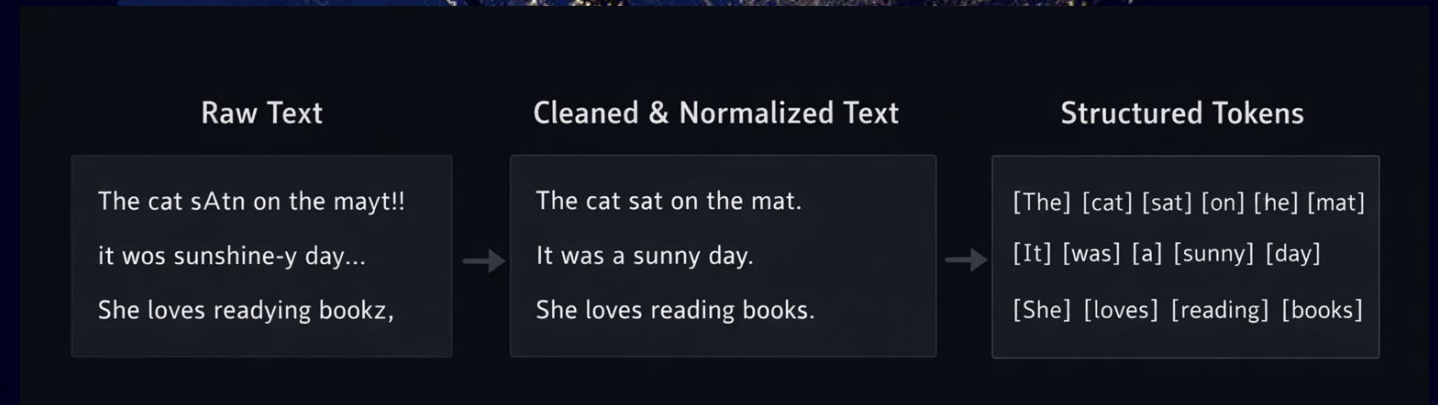
- Text processing is not a preliminary task, but the foundation of all modern NLP systems.
- Before any model can learn, raw human language must be transformed into a structured numerical form.
- This transformation defines what a model can and cannot understand.



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M01 → Chapter 1.1 – Getting Started with Text Processing

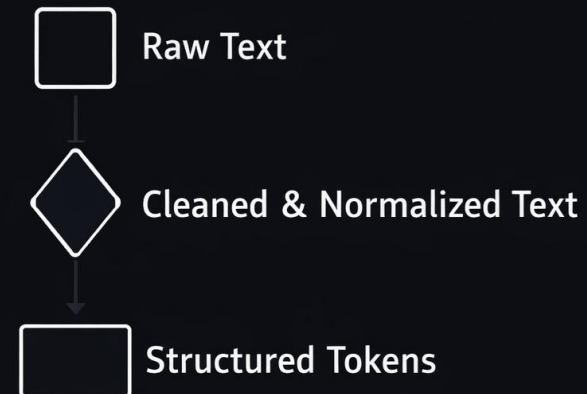
- The goal of preprocessing is to convert raw text into structured, machine-readable units,
- while preserving as much semantic and structural information as possible.
- This step determines the quality and limits of the entire model.



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M01 → Chapter 1.1 – Getting Started with Text Processing

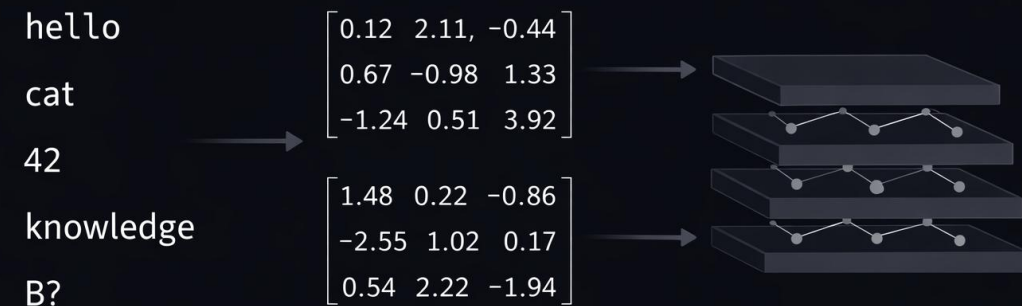
- Cleaning and normalization of noisy text
- Definition of tokens and vocabulary
- Preparation for numerical embedding and vectorization



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M01 → Chapter 1.1 – Getting Started with Text Processing

- Neural networks do not understand symbols or words.
- They operate exclusively on numerical tensors.
- Preprocessing defines how language is translated into numbers, and therefore defines the model's perception of reality.



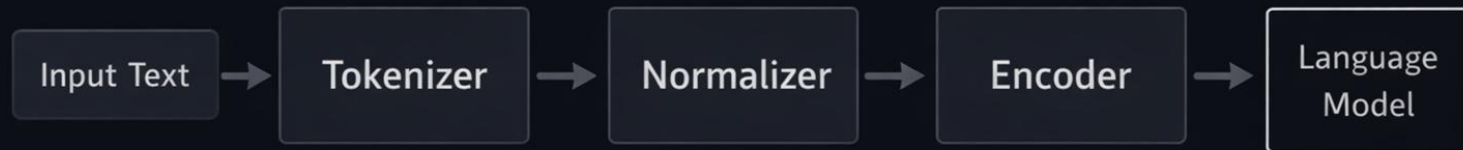


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Chapter 1.2 – Fundamentals of Tokenization



- Modern LLM pipelines rely on highly optimized libraries to ensure efficiency, consistency, and reproducibility.
- These tools define how text becomes training-ready data.

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Chapter 1.2 – Fundamentals of Tokenization

The cat sat
on the mat.



[The]

[cat]

[on]

[the]

[mat]

- Tokenization is the deterministic process of splitting text into discrete units called tokens.
- These tokens are the atomic elements used by language models to represent and process human language.

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Chapter 1.2 – Fundamentals of Tokenization

- Word-level tokenization fails when encountering words that are not present in the training vocabulary.
- These unknown tokens reduce model performance and limit generalization to new data.

I found an obsure word. → <UNK>

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Chapter 1.2 – Fundamentals of Tokenization

- Rule-based tokenizers use predefined patterns to split text while preserving punctuation and structure.
- Regular expressions allow precise control over how symbols, spaces, and contractions are handled.

Hello, Rover! → [Hello] , [,] , [Rover] , [!]

↓
/w+ /
[,]
/!/



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Chapter 1.3 – Byte-Pair Encoding (BPE) and Subword Tokenization

- Subword tokenization was introduced to balance the limitations of word-level and character-level approaches.
- It enables models to represent any word while preserving meaningful linguistic units.

Word-Level Tokens

[] [found] <UNK> []

Subword Tokens

[l] [foun] [d] [ob]
[scure]

Character-Level Tokens

[i] [f] [o] [u] [d] [o] [b] [s] [r]

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Chapter 1.3 – Byte-Pair Encoding (BPE) and Subword Tokenization

Handling the Out-of-Vocabulary (OOV) Problem

- Word-level tokenization fails on unseen or rare words
- Unknown tokens reduce model accuracy and robustness
- Subword tokenization decomposes unknown words into known units
- This guarantees full coverage of any input text

Learning etymological roots.

[et]

[ymo]

[log]

[ical]

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Chapter 1.3 – Byte-Pair Encoding (BPE) and Subword Tokenization

What Is Byte-Pair Encoding (BPE)?



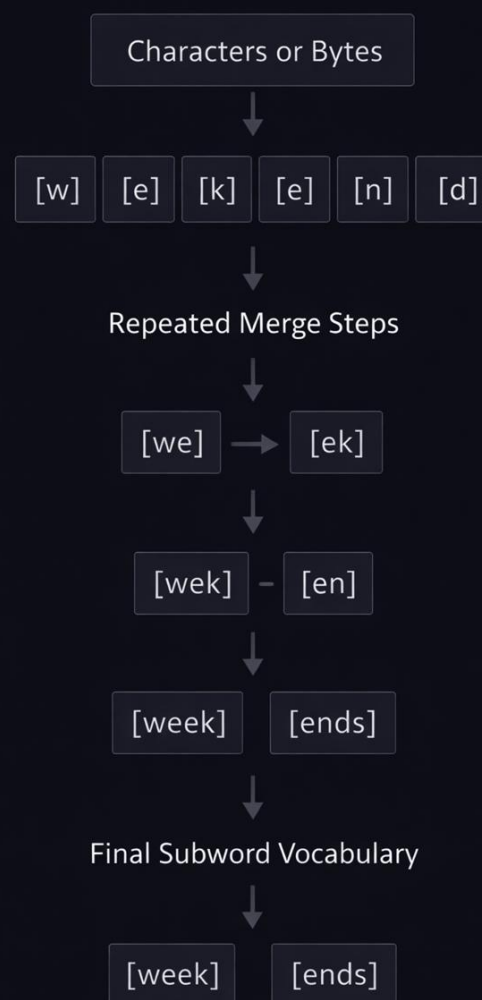
- Byte-Pair Encoding is an iterative subword tokenization algorithm
- It is based on frequency statistics from the training corpus
- The algorithm merges the most frequent adjacent token pairs
- Over time, meaningful subword units are learned

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Chapter 1.3 – Byte-Pair Encoding (BPE) and Subword Tokenization

The BPE Training Process

- Training starts from individual characters or bytes
- All adjacent token pairs are counted across the dataset
- The most frequent pair is merged into a new token
- The process repeats until the target vocabulary size is reached

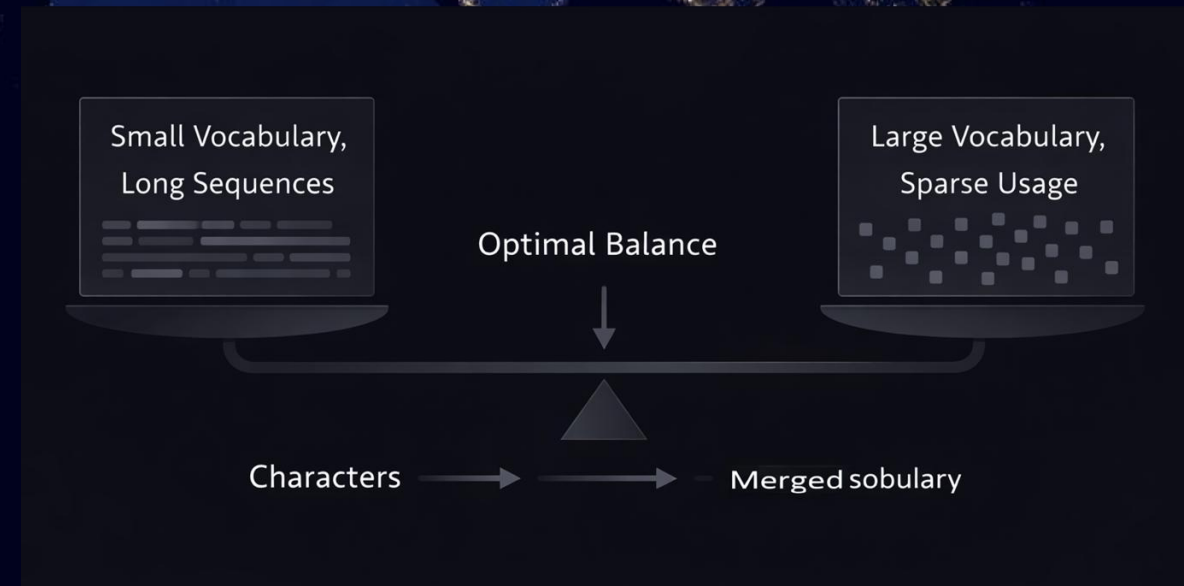


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Chapter 1.3 – Byte-Pair Encoding (BPE) and Subword Tokenization

Vocabulary Size Trade-Offs in BPE

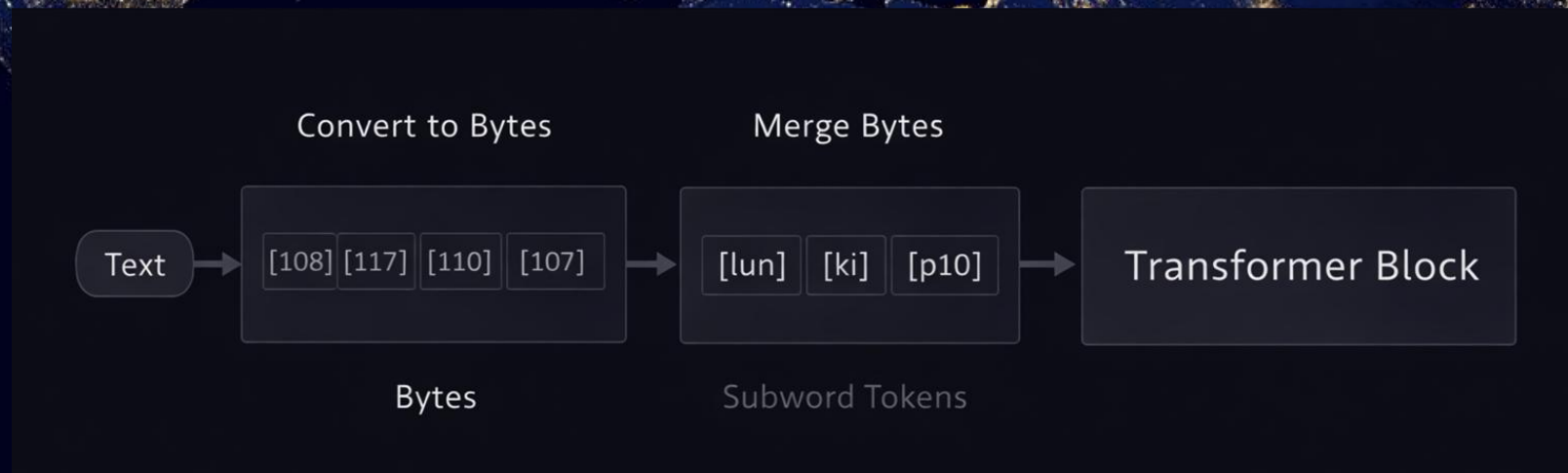
- Small vocabularies increase sequence length
- Longer sequences raise computational cost
- Large vocabularies reduce fragmentation
- Excessively large vocabularies introduce sparsity
- Vocabulary size is a key design decision



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Chapter 1.3 – Byte-Pair Encoding (BPE) and Subword Tokenization

BPE in Transformer and GPT-Style Models



- Modern Transformer models rely on subword tokenization
- Byte-level BPE ensures a fixed and complete base vocabulary
- This approach handles any character or symbol
- It enables efficient and scalable language modeling



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Chapter 1.4 – SentencePiece

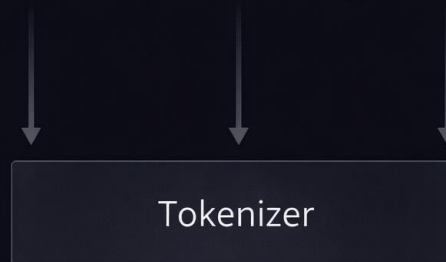
Why SentencePiece Was Introduced

- Traditional tokenizers rely on language-specific rules
- Whitespace-based pre-tokenization fails for many scripts
- SentencePiece processes text as a raw character stream
- This enables fully language-agnostic tokenization

multilingualexamples マルチリンガルサンプル

أمثلة متعددة اللغات

बहुभाषी उदाहरण





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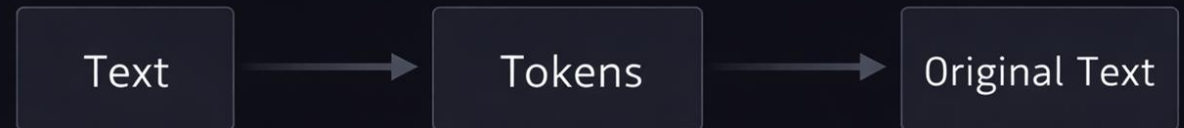
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Chapter 1.5 – Stopwords and Vocabulary Management

Stopwords in Modern NLP Systems

- Stopwords are high-frequency functional words
- Removing them was common in early NLP systems
- Transformer models rely on context and attention
- Modern pipelines often preserve stopwords





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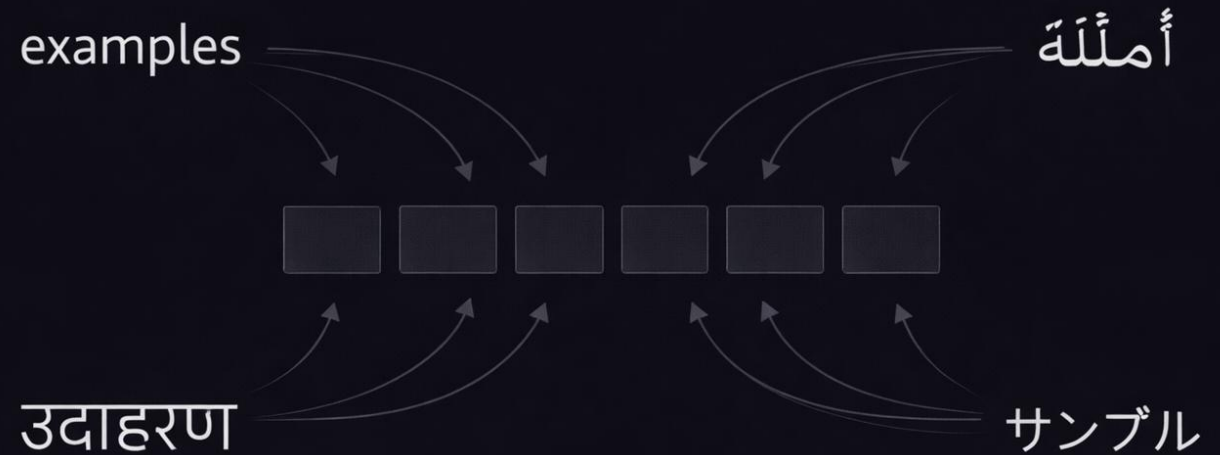
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Chapter 1.6 – Processing Multilingual and Cross-Lingual Text

Challenges of Multilingual Tokenization

- Languages differ in scripts and morphology
- Whitespace is not universal across languages
- Subword tokenization enables shared vocabularies
- This supports cross-lingual and zero-shot learning





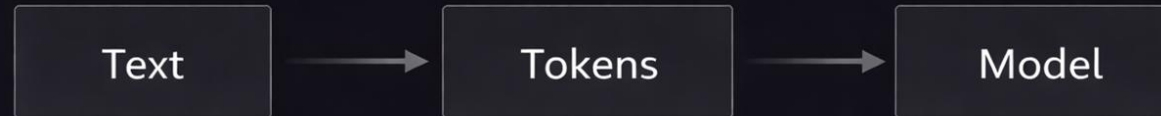
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Chapter 1.7 – Summary and Next Steps

Key Takeaways from Text Preprocessing



- Text preprocessing defines model capabilities
- Tokenization bridges language and mathematics
- Subword methods balance flexibility and efficiency
- Modern systems support multilingual inputs

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M01 – Text Preprocessing

Recommended References and Learning Resources

- Sebastian Raschka – Build a Large Language Model (From Scratch)
- Sennrich et al. (2016) – Neural Machine Translation of Rare Words with Subword Units
- Kudo and Richardson (2018) – SentencePiece: A Language-Independent Tokenizer
- Radford et al. (2019) – Language Models are Unsupervised Multitask Learners
- Hugging Face Course – Tokenization and NLP Fundamentals
- OpenAI Documentation – Tokenizers and tiktoken